

Lecture 11: Euclid's Elements Book I (Part 2)

Proposition 33

A **parallelogram** is a quadrilateral with both pairs of opposite sides parallel. In this proposition, we will show that it suffices to check one pair of opposite sides to be parallel and of equal length. The proof uses SAS.



STRAIGHT lines (——— and ———) which join the adjacent extremities of two equal and parallel straight lines (——— and ———), are themselves equal and parallel.

Draw ——— the diagonal.

———— = - - - - - (hyp.)

△ = △ (pr. 29.)

and ——— common to the two triangles ;

∴ ——— = ———, and △ = △ (pr. 4.) ;

and ∴ ——— || ——— (pr. 27.).

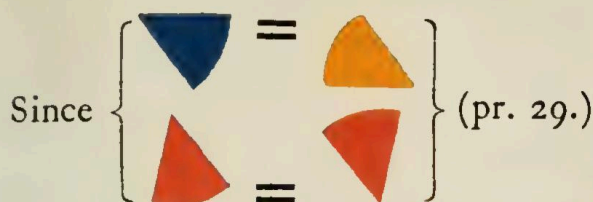
Q. E. D.

Proposition 34

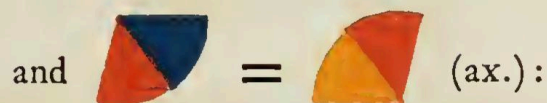
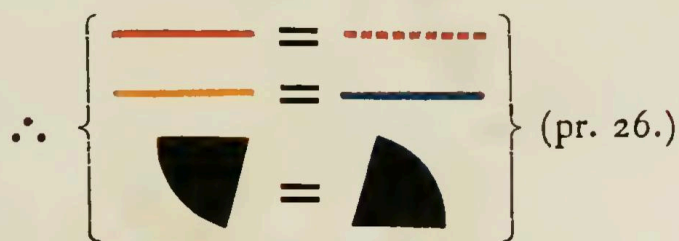
This tells us opposite sides and opposite angles of a parallelogram are equal. The proof uses the previous proposition and the ASA congruence.



THE opposite sides and angles of any parallelogram are equal, and the diagonal (—) divides it into two equal parts.

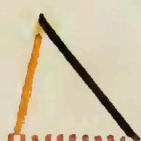


and — common to the two triangles.



Therefore the opposite sides and angles of the parallelo-

gram are equal: and as the triangles



and

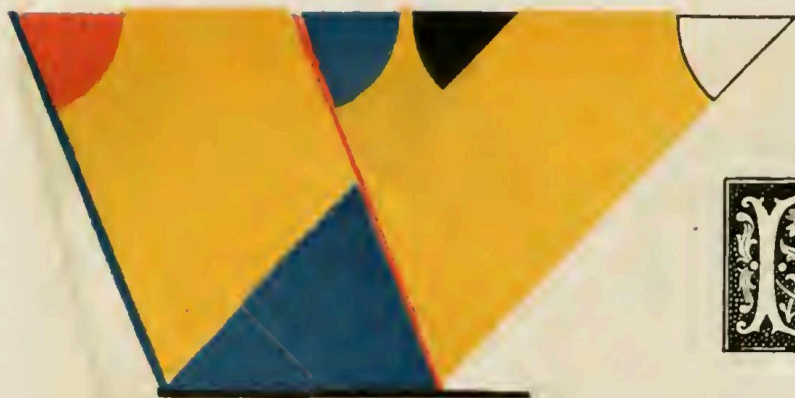


are equal in every respect (pr. 4.), the diagonal divides the parallelogram into two equal parts.

Q. E. D.

Proposition 35

Parallelograms on the same base and between the same parallel lines are equal in area. This is a surprising result because the parallelograms can look very different from each other, yet they enclose the same area. The proof uses ASA. And indeed Euclid has not even introduced the concept of area yet, so this is a purely geometric result about the relative sizes of these parallelograms.



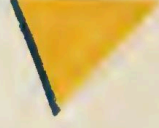
PARALLELOGRAMS
on the same base, and
between the same paral-
lels, are (in area) equal.

On account of the parallels,

$$\begin{array}{lcl}
 \text{red triangle} & = & \text{blue triangle} \\
 \text{black triangle} & = & \text{white triangle} \\
 \text{and } \text{blue line} & = & \text{red line}
 \end{array}
 \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} \text{(pr. 29.)} \\ \text{(pr. 29.)} \\ \text{(pr. 34.)} \end{array}$$

But,  =  (pr. 8.)

$\therefore$  minus  =  ,

and  minus  =  ;

$\therefore$  =  .

Q. E. D.

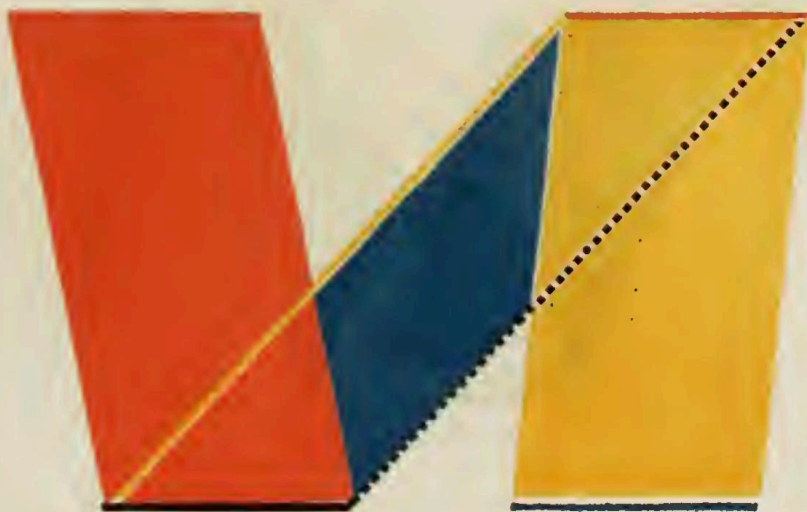
Proposition 36

Parallelograms on equal bases and between the same parallel lines are equal in area. This generalizes Proposition 35 — the base no longer needs to be the same segment, just the same length. The proof reduces to Proposition 35 by connecting the bases.

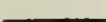


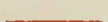
ARALLELO-
GRAMS

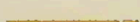
( and ) on
equal bases, and between the
same parallels, are equal.



Draw  and ,

 =  = , by (pr. 34, and hyp.);

$\therefore$  = and $\parallel$ ;

$\therefore$  = and $\parallel$  (pr. 33.)

And therefore  is a parallelogram :

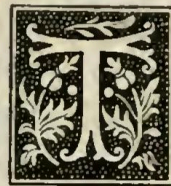
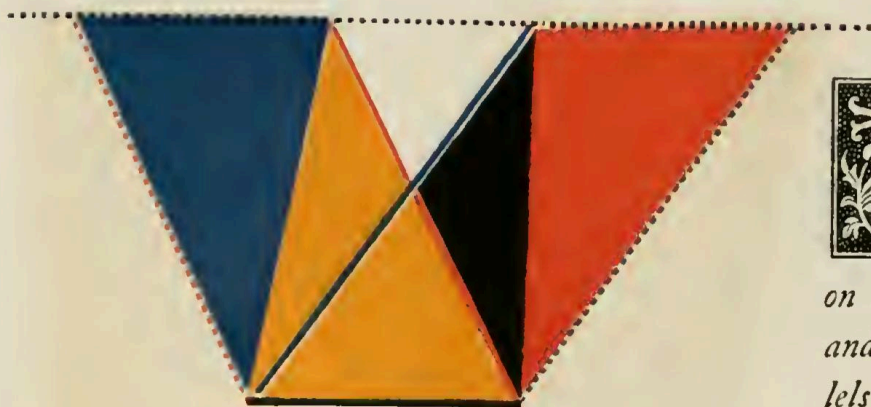
but  =  =  (pr. 35.)

$\therefore$  =  (ax. 1.).

Q. E. D.

Proposition 37

Triangles on the same base and between the same parallel lines are equal in area. This follows immediately from Proposition 35: any triangle is half of a parallelogram on the same base, and Prop 35 tells us those parallelograms are equal.



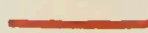
TRIANGLES



and



on the same base (—) and between the same parallels are equal.

Draw  ||  } (pr. 31.)
 ||  }

Produce .



and



are parallelograms

on the same base, and between the same parallels, and therefore equal. (pr. 35.)

$\therefore$ $\left\{ \begin{array}{l} \text{blue triangle} = \text{twice } \text{orange triangle} \\ \text{black triangle} = \text{twice } \text{orange triangle} \end{array} \right\}$ (pr. 34.)

$\therefore$  = .

Q. E. D.

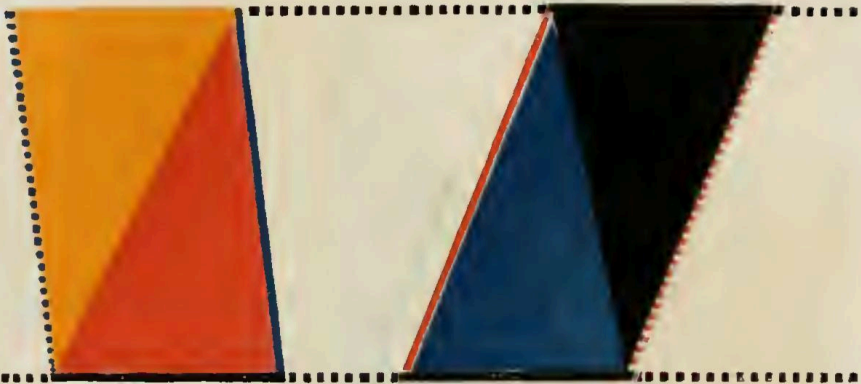
Proposition 38

Triangles on equal bases and between the same parallel lines are equal in area. This is the triangle version of Proposition 36, and follows by the same argument as Proposition 37 using Proposition 36.



TRIANGLES

( and ) on
equal bases and between
the same parallels are equal.



Draw  ||  } (pr. 31.).
and  ||  }

 =  (pr. 36.);

but  = twice  (pr. 34.),

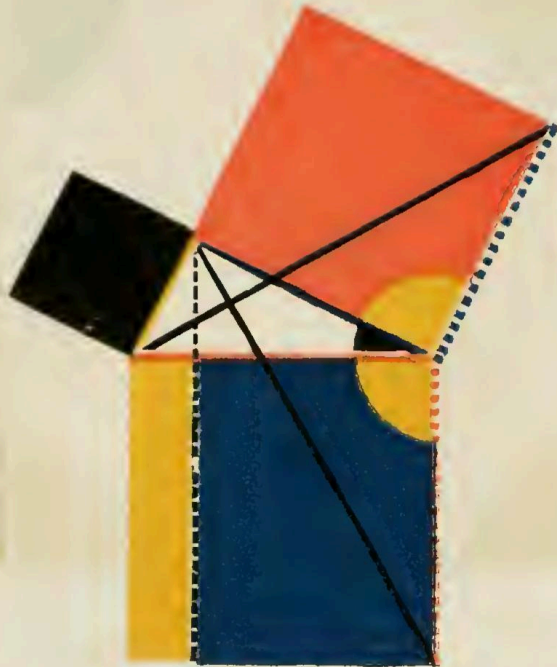
and  = twice  (pr. 34.),

$\therefore$  =  (ax. 7.).

Q. E. D.

Proposition 47 — Theorem (Pythagorean Theorem)

This is one of the most famous theorems in all of mathematics. In a right-angled triangle, the square on the hypotenuse equals the sum of the squares on the other two sides. Euclid's proof is beautiful: he draws the squares on each side and shows the areas match by repeatedly applying congruent triangles and Proposition 41 (parallelogram equals twice a triangle on same base). This is a stunning culmination of Book I.



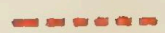
N a right angled triangle
 the square on the
 hypotenuse — is equal to
 the sum of the squares of the sides, (—
 and —).

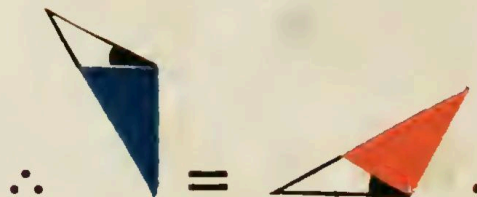
On —, — and —
 describe squares, (pr. 46.)

Draw || - - - - - (pr. 31.)
 also draw — and —.

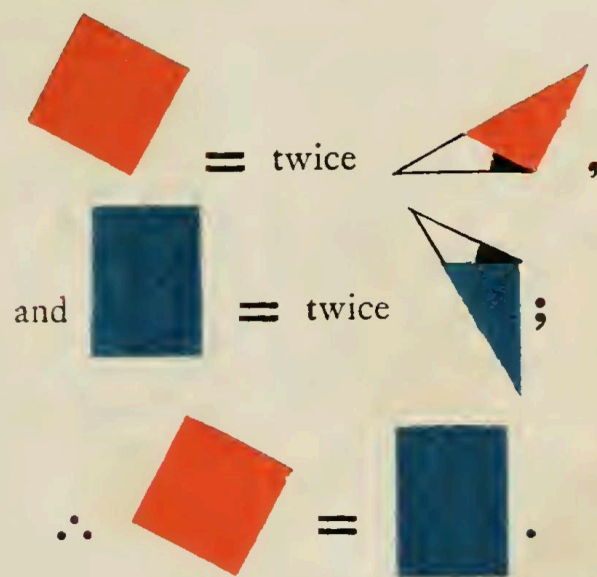


To each add  ∴  =  ,

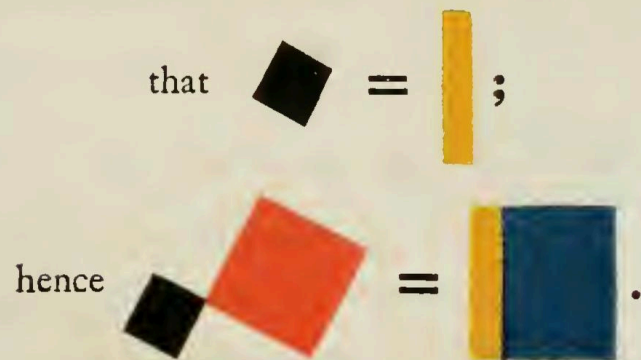
 =  and  =  ;



Again, because — ||



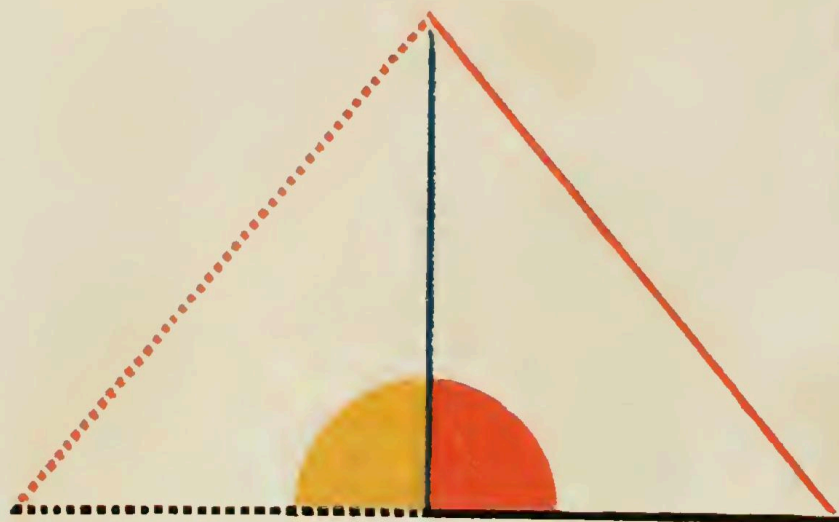
In the same manner it may be shown

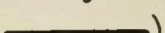


Q. E. D.

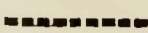
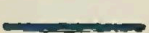
Proposition 48 — Theorem (Converse of Pythagorean Theorem)

If the square on one side of a triangle equals the sum of the squares on the other two sides, then the angle opposite that side is a right angle. This is the converse of Proposition 47, and the proof constructs a right triangle with the same legs and uses SSS congruence to conclude the original triangle must also have a right angle.



IF the square
of one side
() of
a triangle is
equal to the squares of the
other two sides ( and ) , the angle

() subtended by that
side is a right angle.

Draw  $\perp$  and $=$  (prs. 11.3.)
and draw  also.

Since  $=$  (conf.)

$$\text{---}^2 = \text{---}^2;$$

$$\therefore \text{---}^2 + \text{---}^2 = \text{---}^2 + \text{---}^2,$$

$$\text{but } \text{---}^2 + \text{---}^2 = \text{---}^2 \text{ (pr. 47.)},$$

$$\text{and } \text{---}^2 + \text{---}^2 = \text{---}^2 \text{ (hyp.)}$$

$$\therefore \text{---}^2 = \text{---}^2;$$

$$\therefore \text{---} = \text{---};$$

$$\text{and } \therefore \text{---} = \text{---} \text{ (pr. 8.)},$$

consequently  is a right angle.

Q. E. D.